GDX Factor-Residual Reversal Strategy

Research summary: what the code does, what it found, and what to do next
In-sample data 2006-05-22 → 2021-12-31 · Frozen 2026-09-22 at commit `85093c09a2` · repository `sreejaghose/GDX-Factor-Model`, branch `claude/charming-albattani-a06ljy`

1. Executive summary

> **Bottom line.** The pipeline is complete, tested (83 automated tests) and reproducible, and it applied a strict, pre-declared selection protocol to 10,800 strategy configurations. **No configuration met all of the stated requirements on the validation period (2017–2021).** The strategy that was frozen is the best available by the pre-declared tie-break rule and carries the status flag `no_candidate_passed_validation_filters`. The evidence points to a real but thin day-2 reversal effect in GDX's factor residual that is not strong enough, after correcting for the number of configurations tried, to be called a robust edge.

- **Idea.** Explain GDX's daily excess return with SPY/GLD/TLT (rolling OLS). When GDX moves much more or less than its factor exposures imply (a large standardised residual *z*), bet on partial reversal over the next H days.
- **Data reproduced exactly.** The Python pipeline reproduces the workbook's Full-Sample Sweep for all 15 factor subsets (max difference ≈ 1e-10; SPY+GLD: α −3.93 bps/day, β_{SPY} 0.498, β_{GLD} 1.744, R^2 0.6308).
- **Search.** 10,800 configurations (3 factor sets × 5 lookbacks × 4 accumulation windows × 6 thresholds × 5 holding periods × 3 position policies × 2 direction modes), ranked on 2008-05-27 → 2016-12-30 (2,167 trading days), net of 2 bps.
- **Funnel.** Only 10 of 10,800 configurations passed all six development hard filters; all 10 failed at least one filter on validation (every one failed the "still profitable without the best 1% of days" test).
- **Frozen primary:** `GLD|L20|m1|k0.75|H2|A|rev` — GLD-only factor model, 20-day lookback, 1-day residual, |z| ≥ 0.75, hold 2 days, run-out policy, imposed mean reversion. Net Sharpe **0.83** (dev) → **0.46** (validation); gross 0.90 → 0.54; ~6.4 new trades per month; trades every month in both periods.
- **Multiple-testing reality check.** Deflated Sharpe Ratio of the primary = **0.041** (needs ≥ 0.95): with 10,800 trials, a dev Sharpe of about 1.41 would be expected from luck alone. Yet it beats 99.6% of random-timing placebos, and inverting it loses (Sharpe -0.96) — so the timing contains information, just not enough.
- **$1,000 test.** Invested in the frozen primary on 2008-05-23 (fully funded, net of costs), $1,000 grows to ≈ $7,830 by end-2016 and ≈ $13,000 by end-2021, vs $735 buy-and-hold GDX — worst fall −53%. Only the 2017–21 part (+66%, vs +60% for GDX) is free of selection bias.
- **Walk-forward.** Re-applying the selection rule each year 2011–2021 picked 9 different configurations in 11 years and produced a stitched out-of-sample Sharpe of **-0.19** — the selection process itself is unstable.
- **Further experiments (section 8).** After the main run I tested a single-regression version (identical trades), every combination of SPY, IAU, GLD, TLT and GDXJ as factors (165,600 configurations), removing and then re-choosing the hard filters, and a model that picks the best-fitting factor set each day by BIC. **None changed the conclusion**: in every version the median configuration loses in 2017–2021 (≈ −0.22 to −0.27 Sharpe) and the best development configurations do not hold up.
- **Key diagnostic.** The edge sits on **day 2** after the signal, not day 1; the Stage-2 one-day forecast therefore cannot see it (for the primary's settings the estimated slope $\hat{\gamma}$ even points the wrong way), which is why every shortlisted config uses the imposed "reversion" direction.

2. The task and its requirements

Use the GDX factor model in `gold_etf_factor_regression.xlsx` to forecast GDX's next-day return, turn the forecast into a long / short / flat decision, back-test a grid of candidate strategies in-sample, select a small set of robust candidates, freeze them, and ship a runner that applies the frozen strategy to out-of-sample data later with no re-optimisation. Requirements:

- At least one new trade per calendar month.
- Robust to parameter changes — good results must hold in the neighbourhood of a configuration, not at an isolated peak.
- Consistent over time — not driven by one year or a handful of days.
- Signal observed at the close of day t using prices as of that close, and executed at that same close.
- Holding period (exit time, H days) is a tunable parameter like lookback and threshold.

3. Data

- **Prices (input):** daily closes for SPY, IAU, GLD, TLT, GDX, GDXJ; 3,932 rows, 2006-05-22 → 2021-12-31. Zero or blank prices are treated as missing; GDXJ is dropped (not needed; its pre-inception blanks were stored as zeros).
- **FedFunds (input):** monthly effective Fed Funds rate (FRED FEDFUNDS). Daily risk-free = FEDFUNDS/100/252.
- **Returns:** simple close-to-close returns; excess return = return − RF/252. After the first (return-less) row: 3,931 days.
- **Risk-free look-ahead fix.** The workbook maps each day to *that same month's* FEDFUNDS average, which is not known until the month ends. The code implements `rf_mode = same_month` (only to reproduce Excel) and `prior_month` (used everywhere in the strategy). The difference is at most 0.38 bps/day.
- **Adjusted closes.** A heuristic ex-dividend test found no significant dividend-day drops (SPY −20.7 bps vs −45 expected if unadjusted, t = −1.52; TLT −2.9 bps vs −25, t = −0.38), and SPY's 2006 level (87.15 vs ~126 actual) confirms back-adjustment.
- **Reproduction of the workbook.** Full-sample OLS on every one of the 15 factor subsets matches the Full-Sample Sweep sheet (alpha, betas, t-stats, p-values, R^2, adj. R^2, AIC, BIC) within 1e-10, well inside the 1e-4 tolerance. Best BIC: SPY+GLD (−21,304.2); best AIC: SPY+GLD+TLT (β_{TLT} 0.085, t 2.69). IAU and GLD are never used together.

4. What the code does — the model

4.1 Stage 1: rolling factor residual (`src/factor_model.py`)

For factor set F ∈ {GLD, SPY+GLD, SPY+GLD+TLT} and lookback L, at each day t an OLS with intercept is fitted on the window [t−L, t−1] — day t is excluded so today's move cannot contaminate its own benchmark. The out-of-window residual is $e_t = \text{ExRet}_{GDX,t} - \hat{\alpha} - \hat{\beta}^{T}\cdot\text{ExRet}_{F,t}$, the residual volatility $\hat{\sigma}_e$ is the regression standard error $\sqrt{(SSR/(n-p))}$, and the score is $z_t = \Sigma_{j<m}\, e_{t-j} / (\hat{\sigma}_e\sqrt{m})$ for accumulation window m. Rolling sums of X'X and X'y make every window a subtraction with batched solves (≈5 ms per (F, L)); results match a brute-force per-day OLS to 1e-8.

4.2 Stage 2: one-day-ahead forecast (`src/forecast.py`)

At day t, pairs $(z_s, \text{ExRet}_{GDX,s+1})$ for $s \le t-1$ (expanding window, minimum 250 pairs) are used to fit $\text{ExRet}_{s+1} = \gamma\cdot z_s$ without an intercept (so GDX's drift doesn't create a permanent short bias). The newest target is day t's own return, known at t's close. Forecast $\hat{r}_{t+1} = \hat{\gamma}_t\cdot z_t$. The intercept model is kept as a diagnostic, and a rolling W = 500 window was run as a sensitivity check (less stable: $\hat{\gamma}$ negative on 65–89% of days vs ~95% expanding).

4.3 Trade rule and position engine (`src/backtest.py`)

- **Entry:** at close t, if $|z_t| \ge k$, take a position in direction $\text{sign}(\hat{r}_{t+1})$ (*estimated* mode) or $-\text{sign}(z_t)$ (*reversion* mode, imposes mean reversion).
- **Position** is always in {−1, 0, +1}; held from close t to close t+H. Policies: **A** run-out (ignore new signals while in a trade), **B** reset/flip (same-direction signal restarts the clock; opposite flips), **C** reset/flat (opposite closes to flat), and **D** stack-to-2x (reference only, never selectable).
- **Timing:** pos_t is the position after the close of t; strategy return $\text{strat}_{t+1} = \text{pos}_t\cdot\text{ExRet}_{GDX,t+1} - \text{cost}_{t+1}$; cost = 2 bps per unit of position change (entry 1, exit 1, flip 2). Returns are a self-financing 1x overlay in excess of cash.
- The path-dependent loop is compiled with numba (≈0.13 ms per back-test) and unit-tested on hand-built signal sequences for every policy.

5. What the code does — the research protocol

- **Grid** (`config/grid.yaml`): factors {GLD, SPY+GLD, SPY+GLD+TLT} × L {20, 40, 60, 120, 250} × m {1, 2, 3, 5} × k {0.5, 0.75, 1.0, 1.25, 1.5, 2.0} × H {1, 2, 3, 5, 10} × policy {A, B, C} × direction {estimated, reversion} = **10,800** configurations (plus 3,600 D-policy references). The full grid runs in ≈4 s; every configuration's result — losers included — is persisted.
- **One evaluation window for all:** signals are zeroed until the slowest configuration (L = 250, m = 5 plus 250 Stage-2 pairs) has a valid forecast: 2008-05-23. Scoring starts 2008-05-27.
- **Splits** (`config/splits.yaml`): development 2008-05-27 → 2016-12-30 (2,167 days, 103 months) for the full grid; validation 2017-01-03 → 2021-12-31 (1,259 days) for the shortlist only. A logged *validation gate* records every look at validation data and refuses a second, different shortlist unless a reason is logged as a protocol deviation.
- **Hard filters** (development, net of 2 bps): trades in 100% of calendar months; ≥ 150 entries (scaled pro rata for shorter windows); net Sharpe > 0 in both halves; positive net return in ≥ 70% of calendar years; still profitable after removing the best 1% of days; policy ≠ D.
- **Robustness score:** for each configuration, the median development net Sharpe over its neighbourhood (same factors / policy / direction; each of L, m, k, H at its own value or ±1 grid step — up to 81 configurations). Configurations are ranked by this median, not their own Sharpe, so isolated peaks score low. Ties: fewer factors, shorter H, policy B.
- **Statistical checks on the top 10:** Deflated Sharpe Ratio (Bailey & López de Prado) with N = 10,800; 1,000 random-entry placebos with the same trade count, holding lengths and long/short mix; sign-flip; stationary block bootstrap (2,000 reps, mean block 10 days) for a 95% CI. All randomness is seeded.
- **Final pick:** evaluate the 10 on validation once; choose a primary and up to 2 alternates that pass the hard filters on validation, by combined rank of validation Sharpe and neighbourhood score. If none passes, freeze the best by fewest failed filters and flag it.
- **Walk-forward:** for each year Y = 2011…2021, apply the same selection rule to data up to Y−1 and trade the choice in Y.
- **Freeze:** `results/frozen_config.json` (parameters, dev and validation metrics, checks, git hash, data hash) plus a recorded SHA-256 (`6981dc958cd949d4…`) that the OOS runner verifies before running.

5.1 Code map and tests

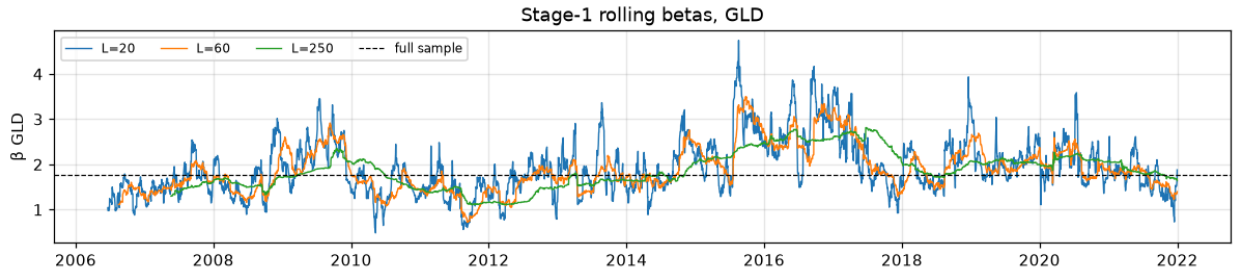
Module	Role
src/data.py	load prices / Fed Funds, returns, rf_mode, adjusted-close check, IS+OOS concatenation with seam checks
src/factor_model.py	full-sample OLS (Excel reproduction), rolling out-of-window residuals, z-scores, caching
src/forecast.py	Stage-2 predictive regression (expanding / rolling), direction modes
src/backtest.py	numba position engine (policies A–D), P&L with costs, trade list
src/grid.py	grid enumeration, common start date, running configs, persistence
src/metrics.py	performance, frequency, consistency, exposure metrics — vectorised over configs
src/splits.py	dev/validation masks, logged validation gate, anchored walk-forward
src/robustness.py	hard filters, neighbourhood score, DSR, placebo, sign-flip, bootstrap, selection rule
src/report.py	event study, heatmaps, ŷ and beta charts, equity, monthly entries, HTML report
scripts/run_is.py	full in-sample workflow: grid → filters → shortlist → validation → freeze → walk-forward → report
scripts/run_oos.py	applies the frozen config to OOS data (hash-verified, no re-optimisation)

83 tests cover: Excel reproduction (15 subsets); brute-force OLS equivalence; Stage-2 vs statsmodels; every position policy on hand-built sequences; cost accounting; metrics vs pandas/statsmodels references; the grid vs single back-tests; a dev-only grid run equal to the dev slice of the full run; the validation gate; walk-forward using only past data; the OOS runner's refusals and seam checks; and the look-ahead checklist — randomising every real price after a date T leaves every z_t, $\hat{r}_{t+1}$ and pos_t for t ≤ T unchanged, and a "cheating" oracle on r_{t+1} scores Sharpe > 20 while the same oracle on r_t does not.

6. Findings

6.1 Stage 1 — the factor model and its stability

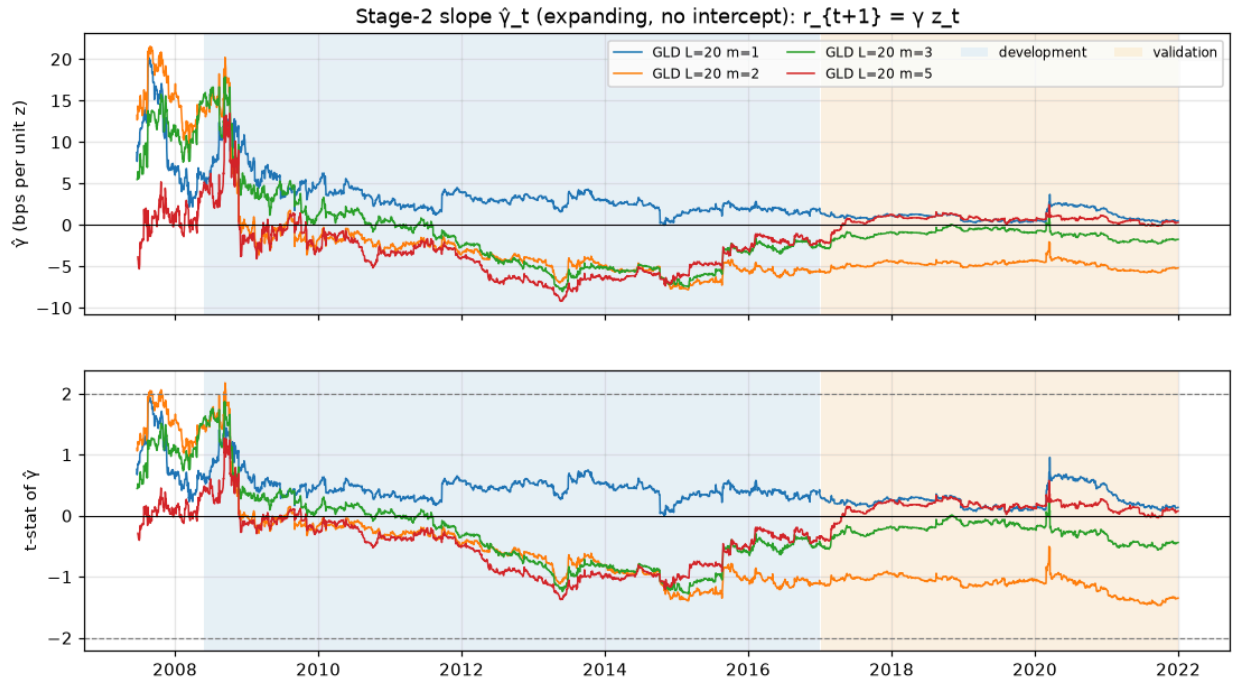
Rolling betas are far from constant: β_{GLD} ranges ≈ 1.0 → 3.5 (peaking 2016), β_{SPY} falls to ≈ 0 in 2019–20, β_{TLT} hovers near zero. This supports refitting on a rolling window. The residual scores are roughly standard normal (SPY+GLD, L = 250: std of z 1.11 for m = 1, 0.99 for m = 5; |z| > 2 on 5–6% of days).



Stage-1 rolling betas for the primary's factor set (GLD), several lookbacks.

6.2 Stage 2 — does the residual predict tomorrow?

Weakly, and not in the way the design assumed. With SPY+GLD, L = 250, the full-IS slope $\hat{\gamma}$ was negative (reversal) for m = 1…5, but significant only for m = 2 (t ≈ −4.2). Decomposing by lag showed why: **yesterday's** residual predicts the next day's return (t = −5.2), today's barely does (t = −0.95), and residuals 2–3 days back predict a small bounce the other way. Raw GDX shows the same pattern (autocorrelation −0.021 at lag 1, −0.069 at lag 2). For the primary's own settings (GLD, L = 20, m = 1), $\hat{\gamma}$ is *positive* throughout — so the "estimated" direction would bet on continuation, while the profitable trade is reversion two days out.



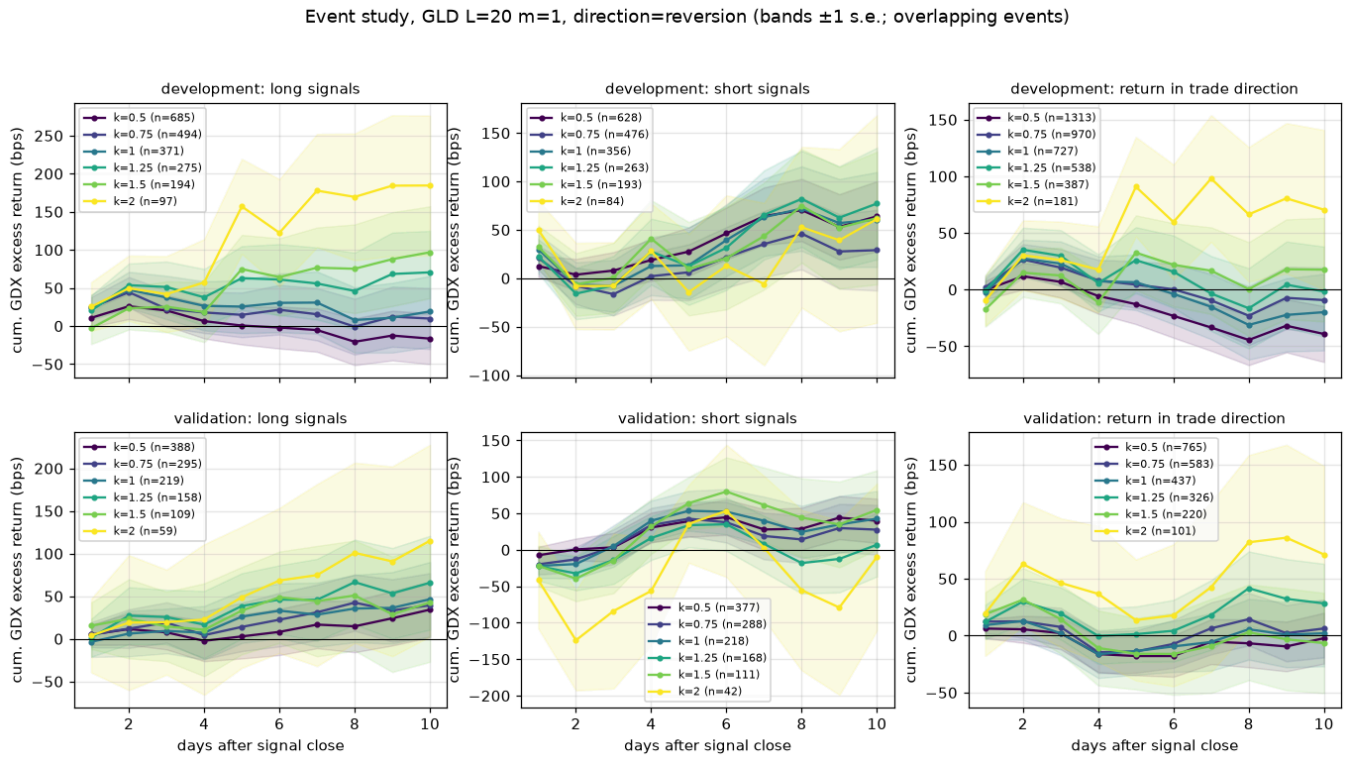
$\hat{\gamma}_t$ and its t-stat over time (GLD, L = 20), development shaded blue, validation orange. The t-stat never exceeds ±2 after 2009.

6.3 Event study — where the edge sits in time

Average cumulative GDX excess return after signals with |z| ≥ k, in the trade's direction (bps):

		day 1	day 2	day 3	day 5	day 10
period	**k**					
development	**0.50**	-0.4	11.8	6.8	-12.9	-39.2
	0.75	2.0	26.5	19.5	4.4	-9.3
	1.00	-2.1	27.4	22.9	6.5	-19.9
	1.25	-0.7	35.2	29.8	26.0	-1.7
	1.50	-17.6	14.9	12.4	32.6	17.6
	2.00	-9.6	30.3	26.1	90.8	70.5
validation	**0.50**	6.2	5.6	2.2	-17.9	-2.2
	0.75	12.3	12.5	7.6	-13.9	6.3
	1.00	9.3	12.9	2.1	-13.6	2.1
	1.25	12.2	29.9	19.6	1.3	28.3
	1.50	18.7	31.6	14.5	-15.9	-6.5
	2.00	19.6	62.7	46.3	13.7	71.2

The day-1 concentration hypothesis is not supported in development: day 1 is ≈ 0 (−18 to +2 bps), the edge appears on **day 2** (+12 to +35 bps) and decays by day 4–10. In validation, day 1 carries roughly half the day-2 total. This is the empirical reason H = 2 is chosen, and the reason a one-day-ahead forecast is aimed at the wrong day.



Event study for GLD, L = 20, m = 1, reversion direction; bands are ±1 s.e. (too narrow — events overlap).

6.4 The whole grid

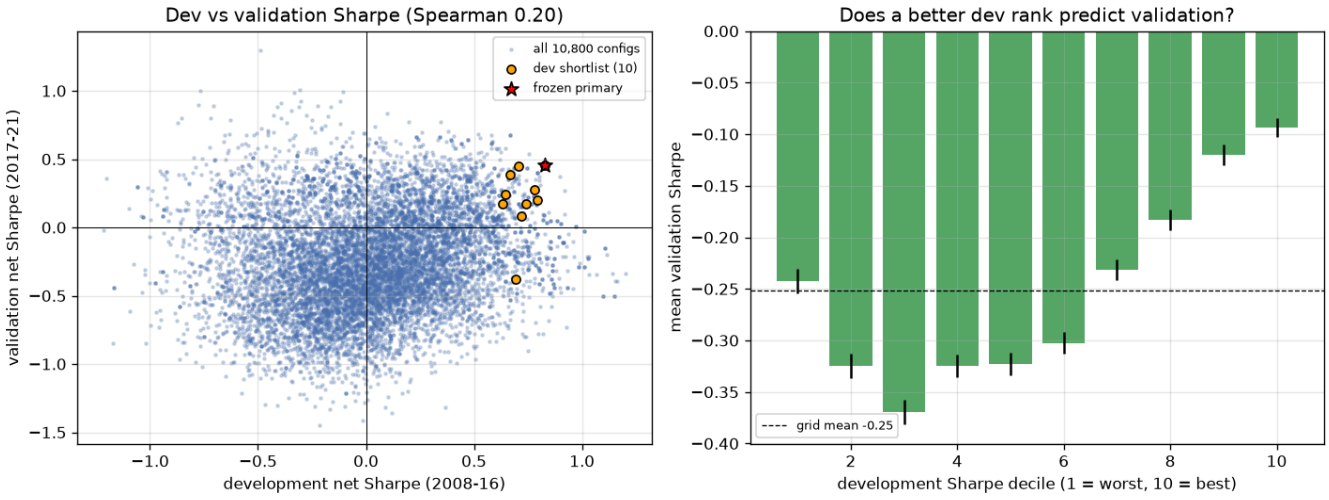
Distribution of net Sharpe (2 bps) across all 10,800 selectable configurations:

	5th pct	25th pct	median	75th pct	95th pct	max	share > 0
development	-0.59	-0.23	0.02	0.27	0.61	1.20	0.52
validation	-0.83	-0.52	-0.27	0.01	0.37	1.30	0.26
full	-0.53	-0.27	-0.07	0.14	0.42	0.72	0.41

The median configuration is roughly break-even in development and **loses in validation** (median −0.27; only 26% of configs positive). Median Sharpe by parameter value shows what "worked" in development did not carry over:

period	development	validation
direction_mode = estimated	-0.090	-0.326
direction_mode = reversion	0.117	-0.217
m = 1	-0.089	0.010
m = 2	0.254	-0.287
m = 3	0.066	-0.235
m = 5	-0.153	-0.478
H = 1	0.119	-0.075
H = 2	0.175	-0.233
H = 3	0.040	-0.344
H = 5	-0.039	-0.407
H = 10	-0.128	-0.235
policy = A_run_out	-0.013	-0.252
policy = B_reset_flip	0.058	-0.274
policy = C_reset_flat	0.020	-0.286
factors = GLD	0.076	-0.322
factors = SPY+GLD	0.017	-0.242
factors = SPY+GLD+TLT	-0.033	-0.238
lookback = 20	-0.037	-0.193
lookback = 40	-0.061	-0.241
lookback = 60	-0.008	-0.270
lookback = 120	0.084	-0.253
lookback = 250	0.112	-0.358
k = 0.5	-0.029	-0.454
k = 0.75	-0.033	-0.373
k = 1.0	0.029	-0.265
k = 1.25	0.075	-0.211
k = 1.5	0.101	-0.227
k = 2.0	0.020	-0.124

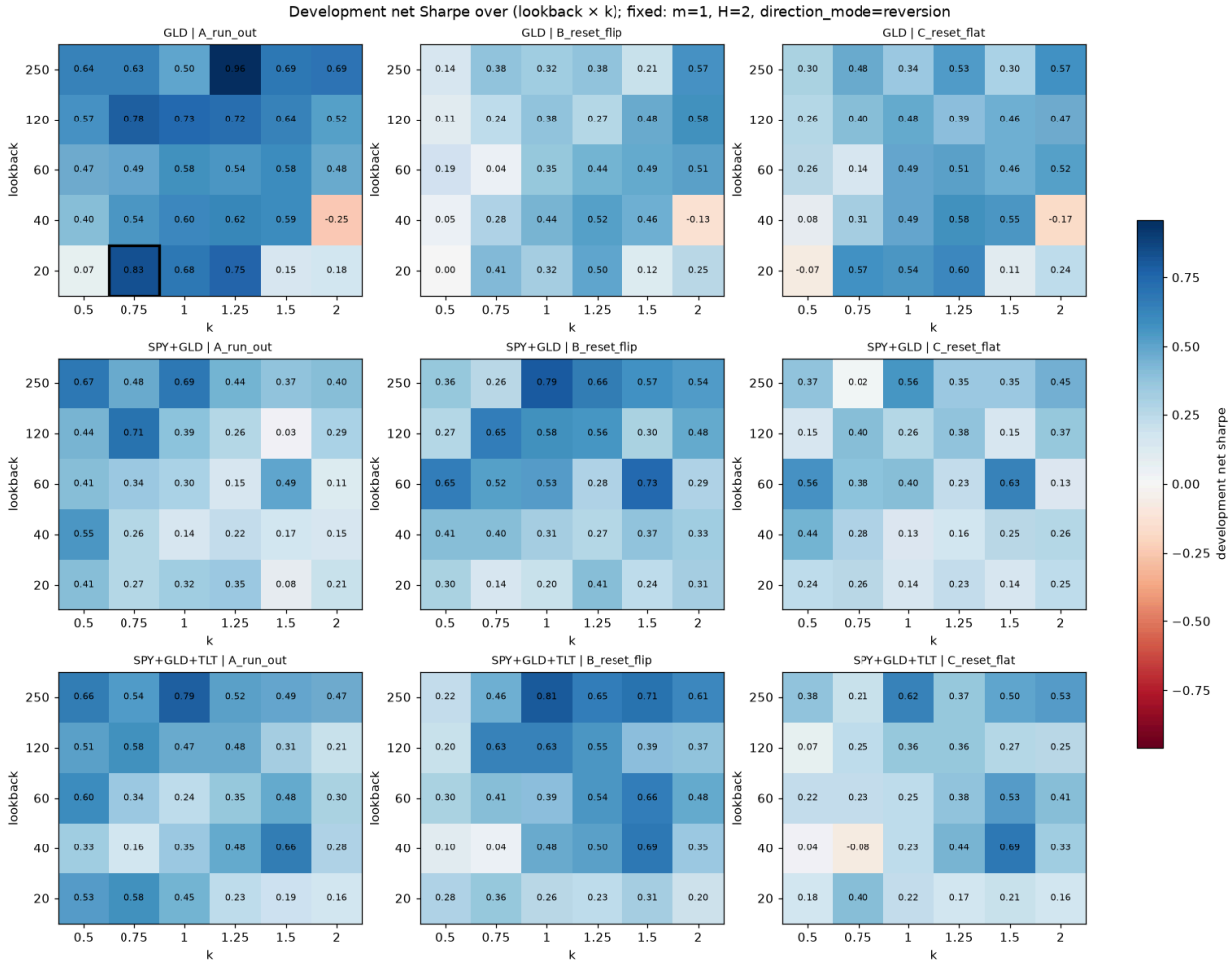
Reversion beat estimated direction in both periods; m = 2 was the best accumulation window in development but among the worst in validation, where only m = 1 is non-negative; short holding periods (H = 1–2) were best in development.



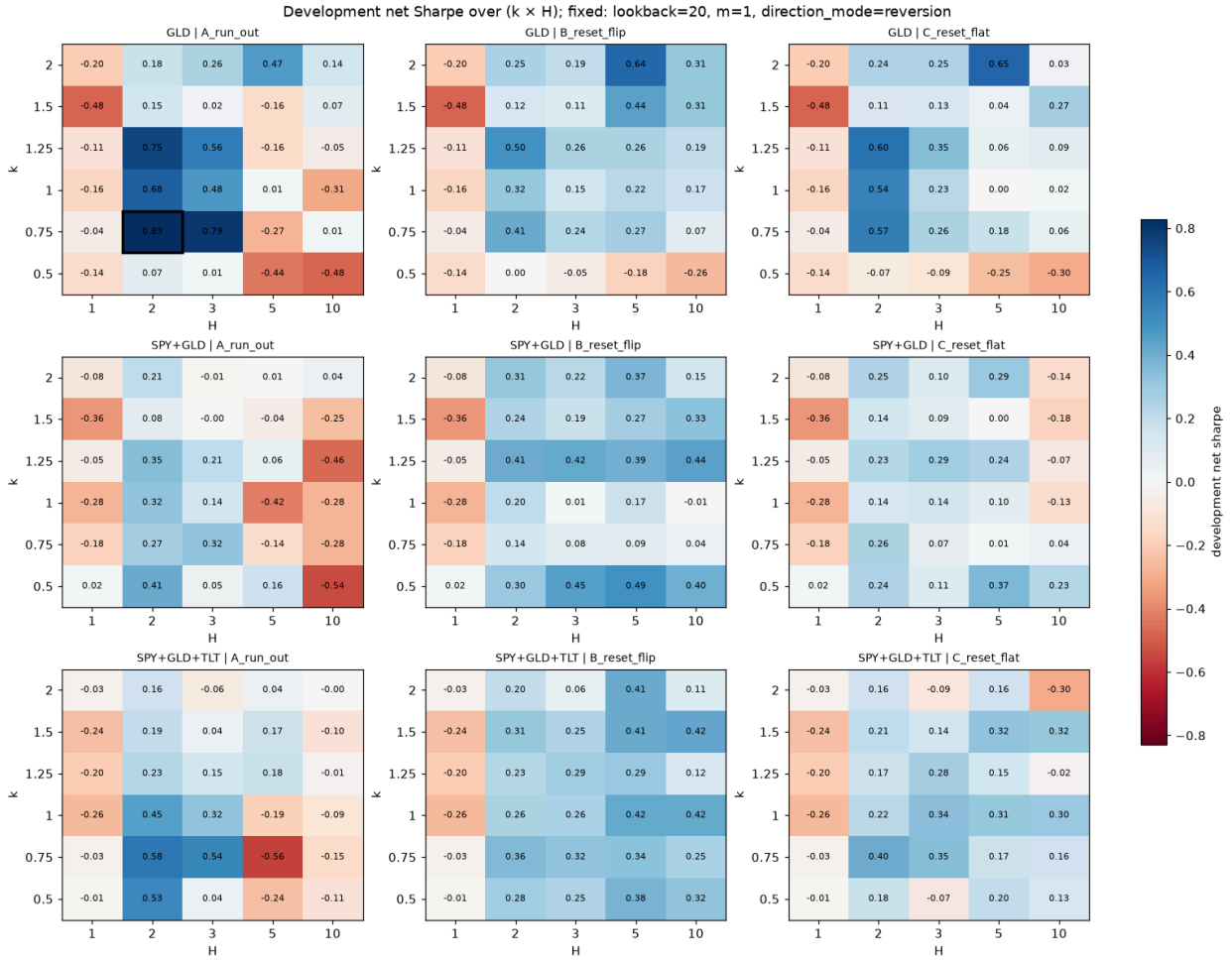
Left: development vs validation Sharpe for every configuration (Spearman rank correlation 0.20). Right: mean validation Sharpe by development-Sharpe decile — better dev ranks help only slightly, and even the top decile averages −0.09 in validation.

6.5 Heatmaps — robustness across neighbours

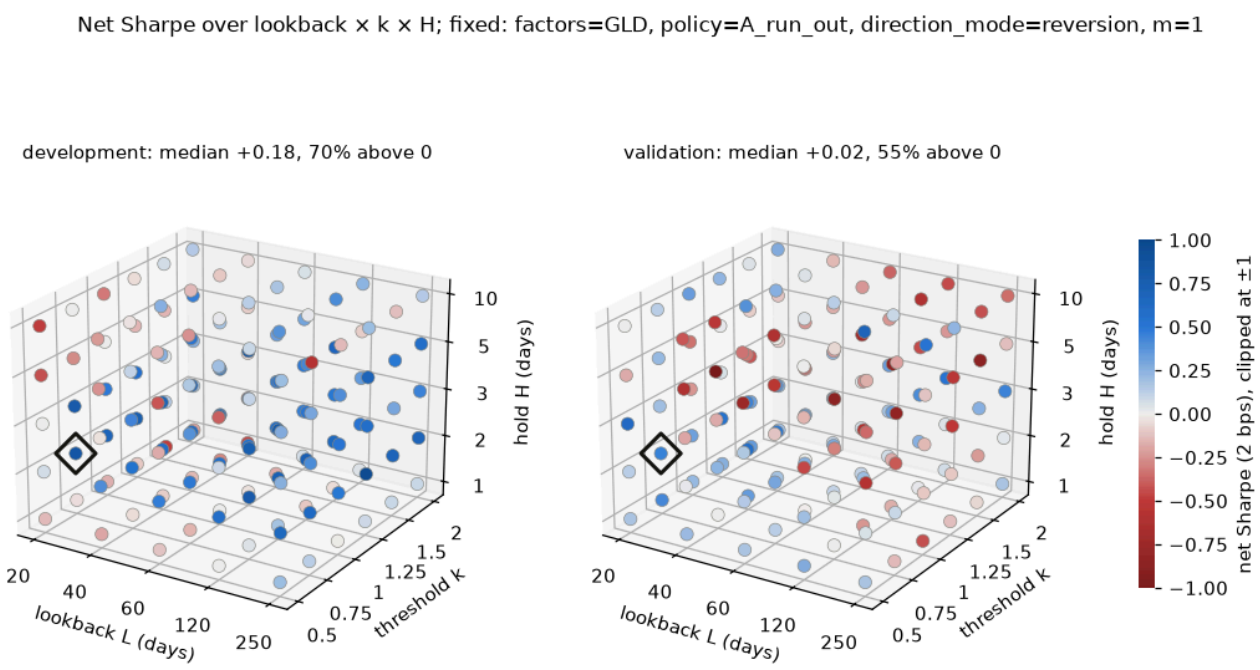
At H = 2, reversion, m = 1, development Sharpe is positive in nearly every (L × k) cell for every factor set and policy (≈ 0.1–0.9), so the day-2 reversal is a broad effect, not an isolated peak. H = 1 is weak or negative almost everywhere, and long holds (H = 5–10) under policy A are frequently negative.



Development net Sharpe over lookback × k (m = 1, H = 2, reversion); rows = factor sets, columns = policies; the primary is outlined.



Development net Sharpe over k × H (L = 20, m = 1, reversion).



The same slice in 3-D (lookback × k × H; GLD, policy A, reversion, m = 1), development vs validation; ◇ marks the primary. In this slice 70% of configurations are positive in development (median +0.18) but only 55% in validation (median +0.02). An interactive version covering every factor set, policy, direction and m is `results/param_cube/param_cube.html`.

6.6 Selection funnel (development)

	configs remaining
cumulative filter	
all selectable configs	10800
trades in 100% of months	2532
≥ 150 entries	2532
Sharpe > 0 in both halves	509
≥ 70% of years positive	183
positive ex top-1% days	10
policy ≠ D (stack)	10

The monthly-trade requirement removes every configuration with k ≥ 1.25 (2,532 survivors all have k ≤ 1.0); positive Sharpe in both halves removes most of the rest; the "ex top-1% days" test is the tightest. Removing the best 1% of days (≈ 22 days) turns the median configuration's annual return to −21%. The best near-misses — failing *only* the monthly-trade filter — score higher on the neighbourhood metric (up to 0.72, e.g. `GLD|L250|m2|k1.25|H1|A|rev`) than anything that passed. **No filter was loosened.**

6.7 Development shortlist and statistical checks

	nbhd median	dev Sharpe	entries	% yrs +	ex-top1% ret	DSR	placebo pct	sign-flip	boot lo	boot hi
config_id										
GLD|L40|m1|k1|H3|A|rev	0.431	0.707	398	0.889	0.003	0.020	99.300	-0.796	0.059	1.355
SPY+GLD|L60|m1|k0.5|H10|B|rev	0.415	0.721	717	0.889	0.034	0.021	99.600	-0.865	0.097	1.378
GLD|L20|m1|k0.75|H2|A|rev	0.408	0.829	655	0.889	0.033	0.041	99.600	-0.962	0.179	1.502
SPY+GLD+TLT|L120|m1|k0.75|H3|B|rev	0.401	0.778	659	0.778	0.042	0.033	99.700	-0.912	0.037	1.575
SPY+GLD|L60|m1|k0.5|H3|B|rev	0.400	0.631	829	0.778	0.002	0.011	98.400	-0.793	0.019	1.269
SPY+GLD|L60|m1|k0.5|H5|B|rev	0.365	0.740	734	0.778	0.041	0.024	98.700	-0.886	0.125	1.383
SPY+GLD+TLT|L120|m1|k0.75|H5|B|rev	0.352	0.647	571	0.778	0.001	0.013	98.700	-0.761	-0.100	1.442
SPY+GLD|L250|m1|k0.5|H2|A|rev	0.332	0.667	775	0.778	0.005	0.015	98.500	-0.798	-0.023	1.381
GLD|L20|m1|k0.75|H3|A|rev	0.311	0.795	504	0.889	0.051	0.035	99.700	-0.887	0.074	1.477
SPY+GLD|L20|m2|k0.5|H10|A|rev	-0.192	0.691	204	0.778	0.016	0.017	98.000	-0.721	0.029	1.394

All ten use the imposed reversion direction and nine use m = 1. Every one beats 98–99.7% of random-timing placebos and loses when inverted, and most bootstrap intervals exclude zero — but every Deflated Sharpe Ratio is 0.01–0.04. The DSR benchmark SR0 is ≈ 1.41 (annualised): with 10,800 configurations whose Sharpe ratios spread widely, a top development Sharpe of 0.6–0.8 is what the best of many noise strategies would show. The configurations are highly correlated, so the effective number of independent trials is smaller than 10,800 and the DSR is conservative — but not by enough to change the conclusion. The tenth shortlisted configuration has a negative neighbourhood median (−0.19) and entered only because exactly ten passed.

6.8 Validation (evaluated once, logged)

	val Sharpe	entries	% months ≥1	% yrs +	Sharpe H1	Sharpe H2	ex-top1% ret	failed filters
config_id								
GLD|L20|m1|k0.75|H2|A|rev	0.459	384	1.000	0.800	0.209	0.629	-0.088	ex_top_days
SPY+GLD|L60|m1|k0.5|H3|B|rev	0.173	417	1.000	0.800	-0.032	0.305	-0.189	halves, ex_top_days
SPY+GLD+TLT|L120|m1|k0.75|H5|B|rev	0.239	284	0.983	0.800	0.529	0.076	-0.167	months, ex_top_days
SPY+GLD+TLT|L120|m1|k0.75|H3|B|rev	0.277	342	0.983	0.600	0.610	0.098	-0.159	months, years, ex_top_days
SPY+GLD|L250|m1|k0.5|H2|A|rev	0.387	439	1.000	0.600	-0.170	0.727	-0.122	halves, years, ex_top_days
SPY+GLD|L60|m1|k0.5|H10|B|rev	0.086	355	0.967	0.800	-0.090	0.194	-0.217	months, halves, ex_top_days
SPY+GLD|L60|m1|k0.5|H5|B|rev	0.172	367	0.983	0.800	-0.025	0.299	-0.188	months, halves, ex_top_days
GLD|L20|m1|k0.75|H3|A|rev	0.199	295	1.000	0.400	-0.290	0.491	-0.167	halves, years, ex_top_days
SPY+GLD|L20|m2|k0.5|H10|A|rev	-0.381	118	1.000	0.200	-0.838	-0.138	-0.351	halves, years, ex_top_days
GLD|L40|m1|k1|H3|A|rev	0.448	232	0.967	0.600	-0.901	1.192	-0.112	months, halves, years, ex_top_days

> No shortlisted configuration passed the hard filters on validation. All ten fail the "ex top-1% days" test (without the best ≈ 13 days they lose 9–35% per year); several also fail positive Sharpe in both halves or the monthly-trade rule.

6.9 The frozen configurations

Primary `GLD|L20|m1|k0.75|H2|A|rev`; alternates `SPY+GLD|L60|m1|k0.5|H3|B|rev` and `SPY+GLD+TLT|L120|m1|k0.75|H5|B|rev` (chosen by fewest validation failures, then combined validation-Sharpe and neighbourhood rank).

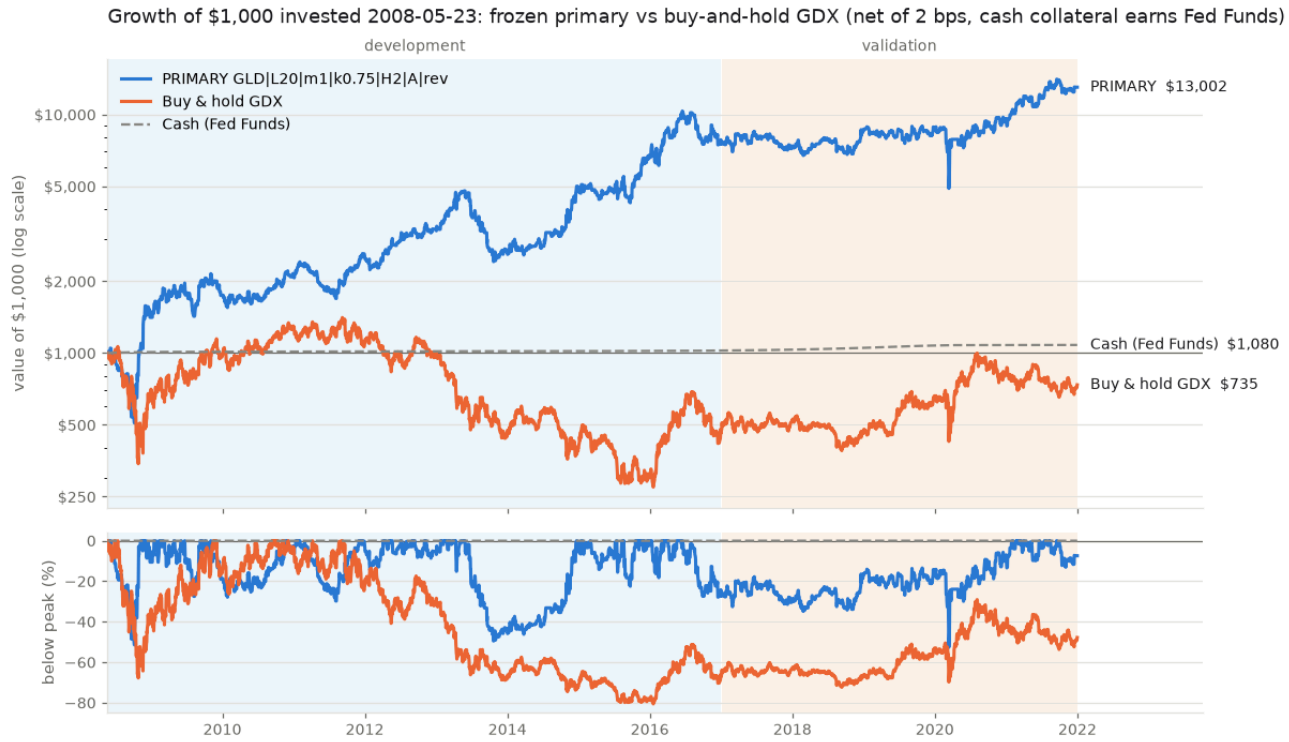
config	GLD|L20|m1|k0.75|H2|A|rev		SPY+GLD|L60|m1|k0.5|H3|B|rev		SPY+GLD+TLT|L120|m1|k0.75|H5|B|rev	
period	**development**	**validation**	**development**	**validation**	**development**	**validation**
net Sharpe	0.829	0.459	0.631	0.173	0.647	0.239
gross Sharpe	0.896	0.539	0.712	0.264	0.704	0.306
ann. return	0.302	0.131	0.278	0.058	0.289	0.078
ann. vol	0.365	0.285	0.440	0.335	0.446	0.329
max DD	-0.687	-0.575	-0.730	-0.849	-0.986	-0.401
NW t	2.409	0.928	1.862	0.384	1.731	0.523
trade hit rate	0.527	0.518	0.569	0.568	0.574	0.567
profit factor	1.299	1.176	1.203	1.058	1.260	1.099
entries	655.000	384.000	829.000	417.000	571.000	284.000
entries/month	6.359	6.400	8.049	6.950	5.544	4.733
% days in mkt	0.604	0.611	0.925	0.912	0.906	0.869
% yrs positive	0.889	0.800	0.778	0.800	0.778	0.800
Sharpe 1st half	0.898	0.209	0.510	-0.032	0.678	0.529
Sharpe 2nd half	0.755	0.629	0.763	0.305	0.614	0.076
ann. ret ex top1% days	0.033	-0.088	0.002	-0.189	0.001	-0.167
corr GDX	0.098	0.134	-0.010	0.012	-0.049	0.033

Statistical checks (development):

	GLD|L20|m1|k0.75|H2|A|rev	SPY+GLD|L60|m1|k0.5|H3|B|rev	SPY+GLD+TLT|L120|m1|k0.75|H5|B|rev
dev net Sharpe	0.829	0.631	0.647
Deflated Sharpe (N=10,800)	0.041	0.011	0.013
SR0 benchmark (ann.)	1.408	1.408	1.408
Prob. Sharpe > 0	0.994	0.968	0.971
placebo percentile	99.600	98.400	98.700
placebo 95th pct Sharpe	0.484	0.456	0.507
sign-flip Sharpe	-0.962	-0.793	-0.761
bootstrap 95% CI low	0.179	0.019	-0.100
bootstrap 95% CI high	1.502	1.269	1.442
skew	1.097	0.296	0.069
kurtosis	17.155	6.692	9.929

The primary trades about 6.4 times a month (655 dev / 384 validation entries), is in the market ≈ 60% of days, holds 2 days on average, turns over ≈ 115–120 units a year, and is nearly uncorrelated with GDX (corr 0.10–0.13), SPY and GLD. Its main weaknesses are its return distribution (kurtosis ≈ 17: the P&L depends on a few large days) and the validation decline (NW t-stat 2.41 → 0.93).

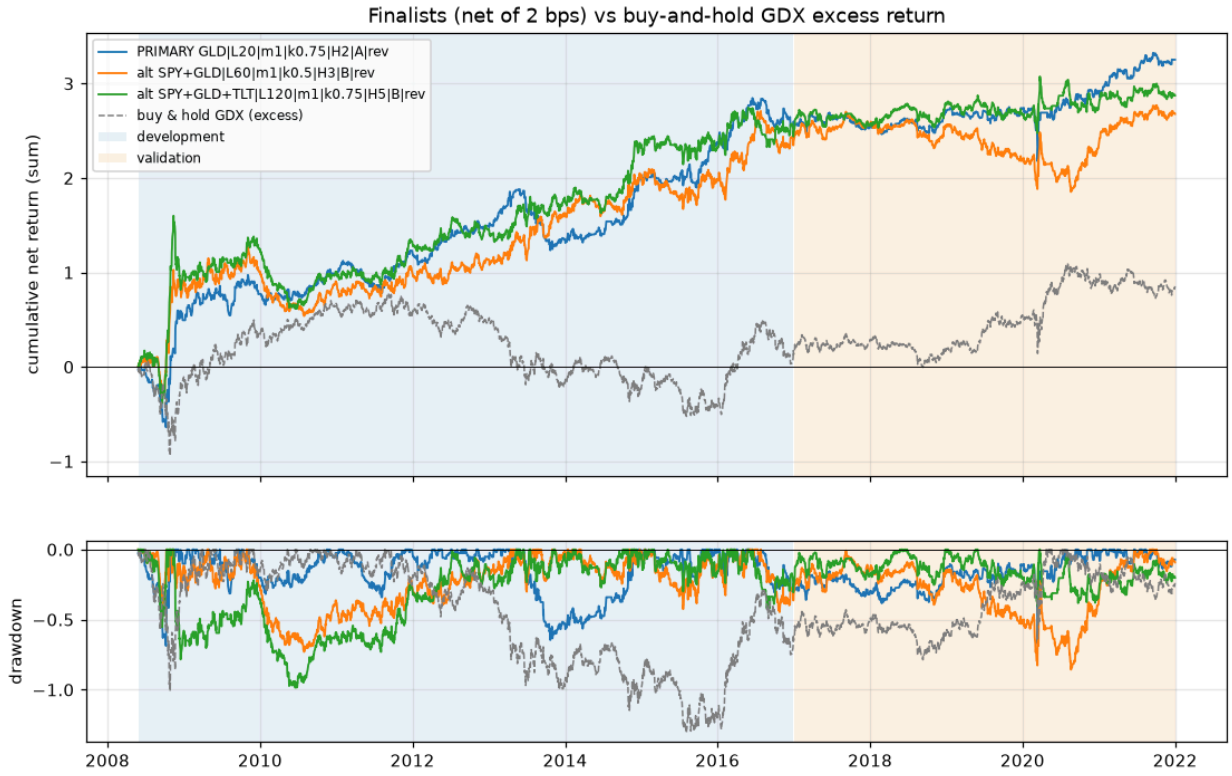
6.10 Growth of $1,000, equity curves, monthly entries and yearly returns



$1,000 invested at the 2008-05-23 close. Fully funded: the cash earns the Fed Funds rate and carries the strategy's 1x long/short GDX position; daily compounding; net of 2 bps per unit traded; no short-borrow fee or taxes.

	start	end of dev (2016)	end of val (2021)	CAGR %	worst drawdown %
PRIMARY: GLD|L20|m1|k0.75|H2|A|rev	1000.00	7830.02	13002.45	20.75	-52.63
Alt: SPY+GLD|L60|m1|k0.5|H3|B|rev	1000.00	4856.83	5192.55	12.87	-65.32
Alt: SPY+GLD+TLT|L120|m1|k0.75|H5|B|rev	1000.00	5190.91	6209.86	14.36	-69.57
Buy & hold GDX	1000.00	459.24	735.10	-2.24	-80.53
Cash (Fed Funds)	1000.00	1022.78	1080.32	0.57	0.00

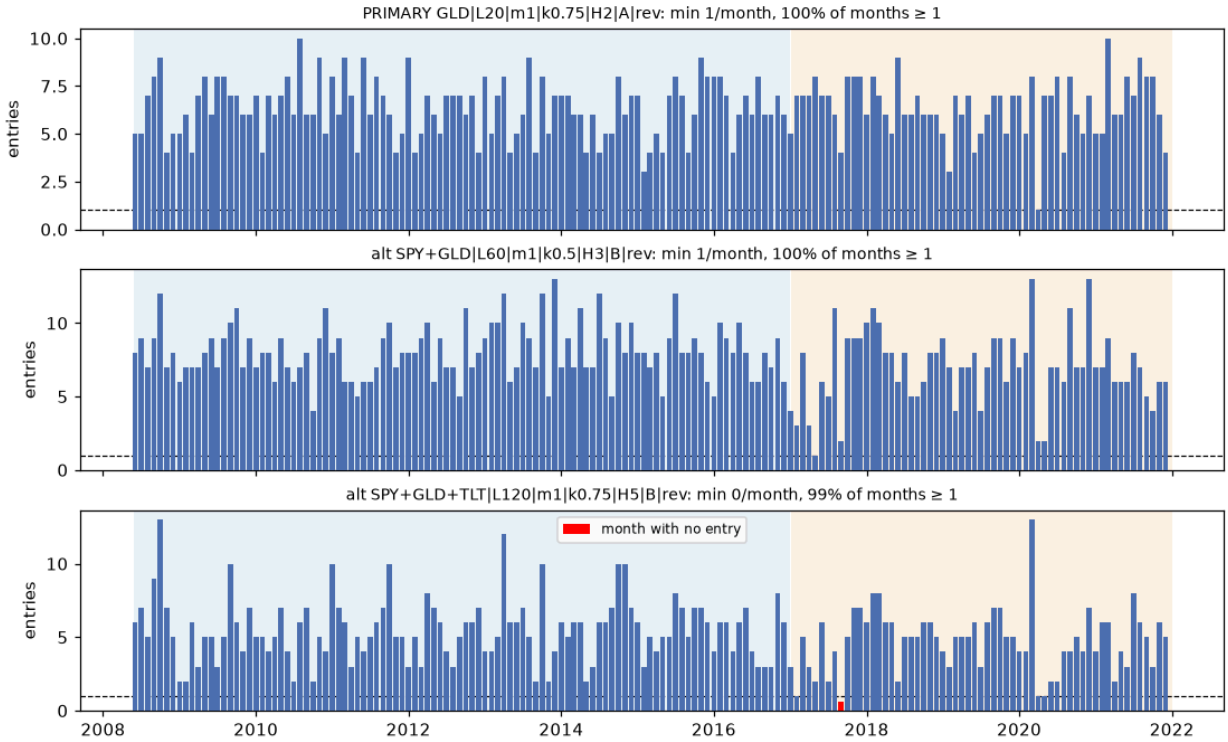
The primary turns $1,000 into ≈ $7,830 by the end of development and ≈ $13,000 by the end of 2021 (≈ 20.7% a year), against $735 for buy-and-hold GDX and $1,080 in cash — but with a worst fall of −53% from peak (GDX: −81%). **Caution:** the development segment is in-sample (the configuration was chosen on it), so the ≈ 7.8× there is inflated by selection. The fair view is validation only: $7,830 → $13,002 (+66% over five years, ≈ 10.7% a year), while buy-and-hold GDX went $459 → $735 (+60%) over the same five years.



Cumulative net P&L (1x notional, additive) and drawdowns vs buy-and-hold GDX excess return.

	PRIMARY	Alt 1	Alt 2	Buy&hold GDX
year				
2008	0.494	0.912	1.021	-0.107
2009	0.279	0.093	0.222	0.443
2010	0.235	-0.108	-0.234	0.336
2011	0.218	0.109	0.314	-0.119
2012	0.268	0.131	0.136	-0.050
2013	-0.143	0.491	0.309	-0.672
2014	0.667	0.411	0.586	-0.059
2015	0.342	-0.066	-0.029	-0.188
2016	0.242	0.415	0.155	0.546
2017	-0.072	0.168	0.073	0.137
2018	0.189	0.006	0.242	-0.076
2019	0.023	-0.357	-0.124	0.350
2020	0.186	0.149	0.092	0.373
2021	0.328	0.324	0.109	-0.061
dev ann. Sharpe	0.829	0.631	0.647	0.032
val ann. Sharpe	0.459	0.173	0.239	0.413

Yearly net returns (sums of daily returns, 1x notional). The primary had 12 of 14 years positive; validation years: 2017 −7.2%, 2018 +18.9%, 2019 +2.3%, 2020 +18.6%, 2021 +32.8%. Buy-and-hold GDX has a validation Sharpe of 0.41 — close to the primary's 0.46.



New entries per calendar month. The primary and alternate 1 trade every month in both periods; alternate 2 misses one month in 2017.

6.11 Does it depend on 2008 / 2020? Does it survive costs?

	all years		ex 2008		ex 2020		ex 2008 & 2020	
	Sharpe	**ann. return**	**Sharpe**	**ann. return**	**Sharpe**	**ann. return**	**Sharpe**	**ann. return**
PRIMARY	0.709	0.239	0.695	0.212	0.749	0.244	0.745	0.215
Alt 1	0.487	0.197	0.364	0.136	0.512	0.201	0.379	0.135
Alt 2	0.519	0.211	0.389	0.143	0.558	0.221	0.420	0.147

The primary does **not** depend on 2008 or 2020 (Sharpe 0.71 all years; 0.70 ex 2008; 0.75 ex 2020; 0.75 ex both). The two alternates lean on 2008 (≈ 0.5 → ≈ 0.37 without it).

	development	validation
gross (0 bps)	0.90	0.54
2 bps	0.83	0.46
5 bps	0.73	0.34

The edge survives 5 bps per unit traded (validation Sharpe 0.34), but each basis point removes a visible share, and with ≈ 115–120 units of turnover a year, costs of 2 bps take ≈ 2.4% a year. Costs are not the binding problem; the thinness and tail-dependence of the edge are.

6.12 Walk-forward of the selection rule

	config chosen (data to Y-1)	train Sharpe	test-year Sharpe	entries
test year				
2011	GLD|L250|m1|k0.5|H2|A|rev	1.18	1.48	99
2012	GLD|L250|m1|k0.5|H2|A|rev	1.21	-0.58	81
2013	GLD|L250|m1|k0.5|H3|A|rev	1.09	-1.14	70
2014	GLD|L250|m1|k0.5|H2|A|rev	0.87	0.08	85
2015	GLD|L120|m2|k0.5|H2|A|rev	0.87	-0.51	86
2016	SPY+GLD|L120|m1|k0.75|H3|B|rev	0.74	0.16	71
2017	GLD|L40|m1|k1|H3|A|rev	0.71	-1.78	42
2018	GLD|L20|m1|k0.75|H3|A|rev	0.72	-0.35	58
2019	GLD|L20|m1|k0.75|H2|A|rev	0.75	0.11	69
2020	SPY+GLD+TLT|L250|m2|k1|H1|A|rev	0.97	-0.85	49
2021	SPY+GLD|L60|m1|k0.5|H3|B|rev	0.45	1.20	77

9 distinct configurations in 11 years, 2.7 of 7 parameters changing per switch (red flag raised); only the reversion direction was stable. Test-year Sharpe was negative in 6 of 11 years despite training Sharpe of 0.45–1.2 every year; stitched out-of-sample Sharpe **-0.19** (-6.3% a year). The primary itself was the walk-forward choice in only 1 year (2019).

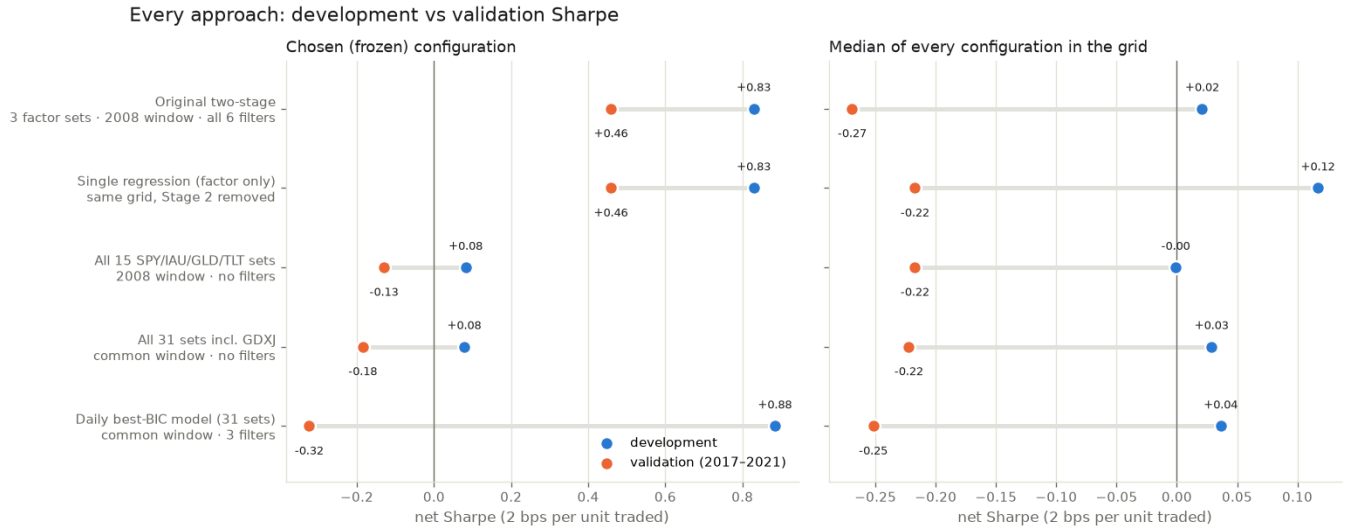
7. Out-of-sample runner (ready, not yet run on real OOS data)

```
python scripts/run_oos.py --is data/raw/gold_etf_factor_regression.xlsx --oos data/raw/<OOS file>.xlsx \
    --config results/frozen_config.json --out results/oos/
```

- Refuses to run if `frozen_config.json`'s SHA-256 differs from the hash recorded at freeze, or if the IS workbook is not the one the config was frozen on.
- Concatenates IS and OOS prices with seam checks: identical overlapping rows are dropped, differing overlaps or a gap of more than 4 weekdays are refused.
- Uses IS history as warm-up so rolling betas and $\hat{y}$ are live on OOS day 1 and keep updating (part of the frozen rule); no grid parameter changes. Frozen configs start flat and may first trade at the last IS close.
- Reports OOS-only metrics, hard-filter pass/fail, Sharpe at 0/2/5 bps, with/without 2008 and 2020, all Step-10 diagnostics, and an IS-vs-OOS side-by-side table; the run manifest records a check that the frozen development Sharpe still reproduces (currently within 1e-6).

8. Further experiments (after validation was seen)

> Everything in this section was specified after the 2017–2021 validation results were known, so every validation number here is a logged protocol deviation and only fresh out-of-sample data can confirm any of it. The original frozen strategy was left untouched throughout.



Left: the configuration each approach selected. Right: the median across all of that approach's configurations. Every grid's median falls from ≈ 0 in development to ≈ −0.22 to −0.27 in validation.

	configs	configs tried so far (DSR N)	development window	chosen config	chosen: dev Sharpe	chosen: val Sharpe	grid median dev	grid median val	share of grid > 0 in val	walk-forward Sharpe	DSR
approach											
Original two-stage — 3 factor sets · 2008 window · all 6 filters	10800	10800	2008-05-27 → 2016-12-30	GLD|L20|m1|k0.75|H2|A|rev	0.83	0.46	0.02	-0.27	0.26	-0.19	0.04
Single regression (factor only) — same grid, Stage 2 removed	5400	16200	2008-05-27 → 2016-12-30	GLD|L20|m1|k0.75|H2|A|rev	0.83	0.46	0.12	-0.22	0.30	-0.19	0.08
All 15 SPY/IAU/GLD/TLT sets — 2008 window · no filters	54000	54000	2008-05-27 → 2016-12-30	GLD|L250|m1|k1.25|H1|A|rev	0.08	-0.13	-0.00	-0.22	0.28	0.22	0.00
All 31 sets incl. GDXJ — common window · no filters	111600	165600	2011-10-10 → 2016-12-30	TLT|L250|m1|k1.25|H1|B|est	0.08	-0.18	0.03	-0.22	0.27	-0.14	0.00
Daily best-BIC model (31 sets) — common window · 3 filters	3600	169200	2011-10-10 → 2016-12-30	BEST_BIC|L250|m3|k0.5|H5|B|est	0.88	-0.32	0.04	-0.25	0.24	0.45	0.09

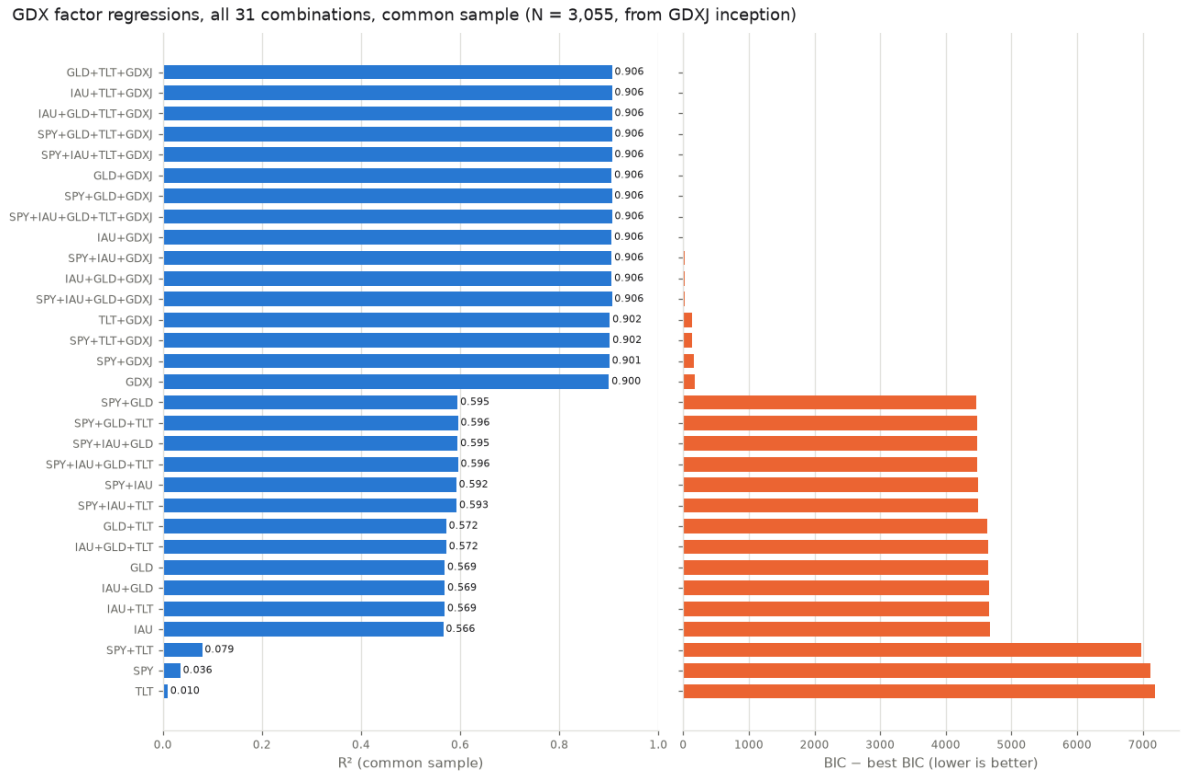
8.1 One regression instead of two

Dropping the Stage-2 forecast (trading −sign(z) directly from the factor regression) gives **identical trades and results** for the configurations that were selected: they all used the imposed "reversion" direction, which never looks at $\hat{y}$. The only practical gains are simplicity and 250 fewer days of warm-up.

8.2 Every factor combination (SPY, IAU, GLD, TLT, GDXJ)

All 31 combinations were fitted on the common sample from GDXJ's inception (Nov 2009). Models that include GDXJ explain ≈ 90–91% of GDX's daily moves against at most ≈ 60% without it, so with GDXJ the residual measures GDX against junior

miners rather than against gold. IAU and GLD together are near-perfectly collinear (VIF ≈ 320): their betas split arbitrarily while the fit barely changes.



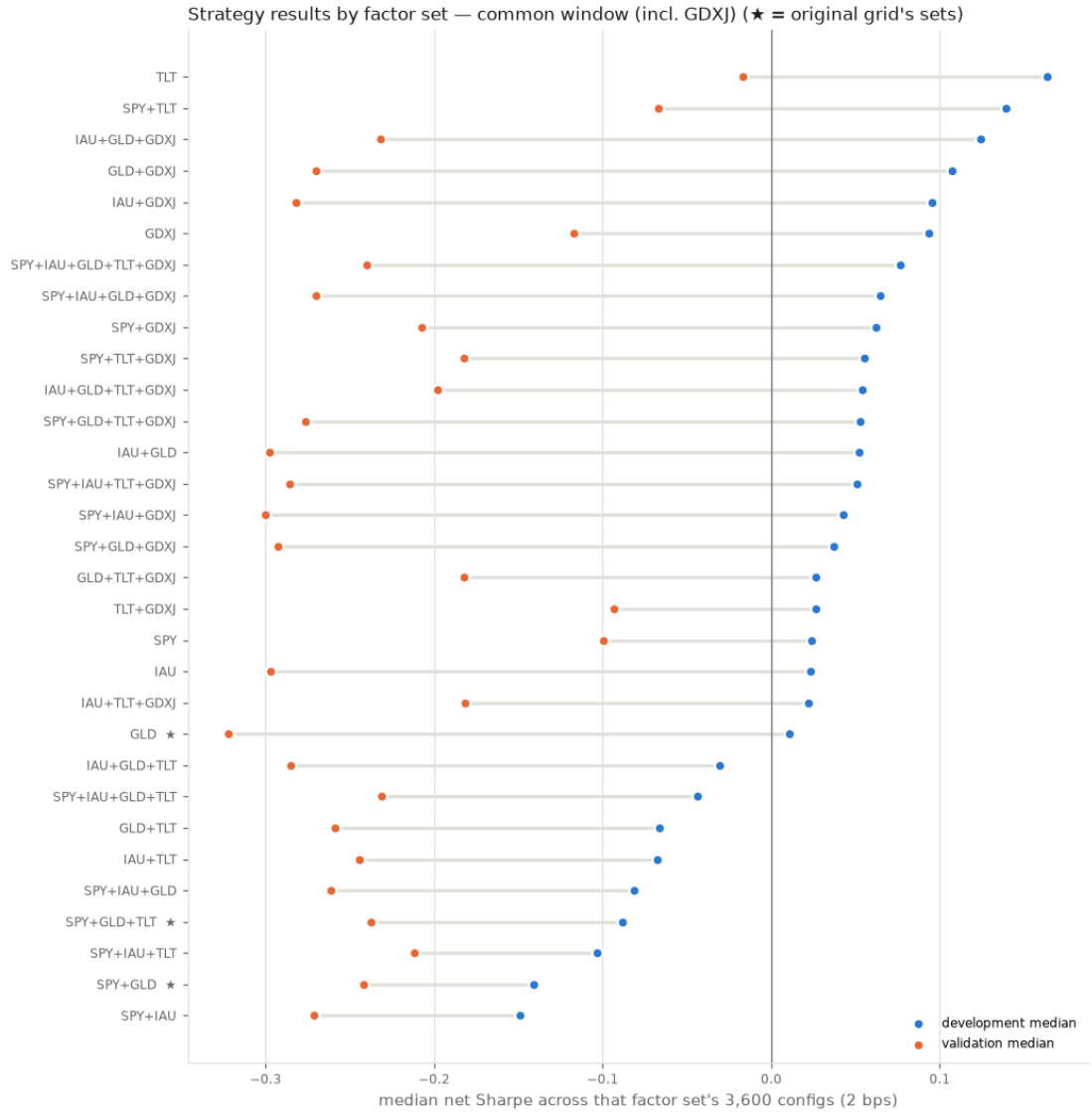
R² and BIC for all 31 combinations, same sample for all.

	R2	BIC	BIC_rank	Beta_GLD	Beta_IAU	Beta_GDXJ	Beta_SPY	Beta_TLT	max_vif
Factors									
GLD+TLT+GDXJ	0.906	-21357.751	1	0.243	–	0.711	–	0.067	2.363
IAU+TLT+GDXJ	0.906	-21355.069	2	–	0.239	0.712	–	0.068	2.346
IAU+GLD+TLT+GDXJ	0.906	-21350.392	3	0.437	-0.194	0.711	–	0.067	323.521
SPY+GLD+TLT+GDXJ	0.906	-21350.152	4	0.241	–	0.712	-0.010	0.062	2.455
SPY+IAU+TLT+GDXJ	0.906	-21347.466	5	–	0.238	0.713	-0.010	0.063	2.442
GLD+GDXJ	0.906	-21345.881	6	0.264	–	0.707	–	–	2.231
SPY+GLD+GDXJ	0.906	-21344.770	7	0.253	–	0.712	-0.034	–	2.455
SPY+IAU+GLD+TLT+GDXJ	0.906	-21342.815	8	0.438	-0.197	0.712	-0.010	0.061	323.558

Best models without GDXJ:

	R2	BIC	BIC_rank	Beta_GLD	Beta_SPY	Beta_TLT	max_vif
Factors							
SPY+GLD	0.595	-16889.399	17	1.794	0.363	–	1.001
SPY+GLD+TLT	0.596	-16886.143	18	1.778	0.393	0.077	1.349
SPY+IAU+GLD	0.595	-16882.478	19	2.312	0.362	–	320.954
SPY+IAU+GLD+TLT	0.596	-16879.075	20	2.261	0.393	0.076	321.696

Every combination was then run through the strategy grid (3,600 configurations each; 54,000 on the 2008 window without GDXJ and 111,600 on a common window from Oct 2011 with GDXJ). **No factor set has a positive median validation Sharpe**; the least bad are TLT and SPY+TLT (≈ −0.02 and −0.07), which explain almost nothing of GDX, so their "residual" is just GDX's own return.

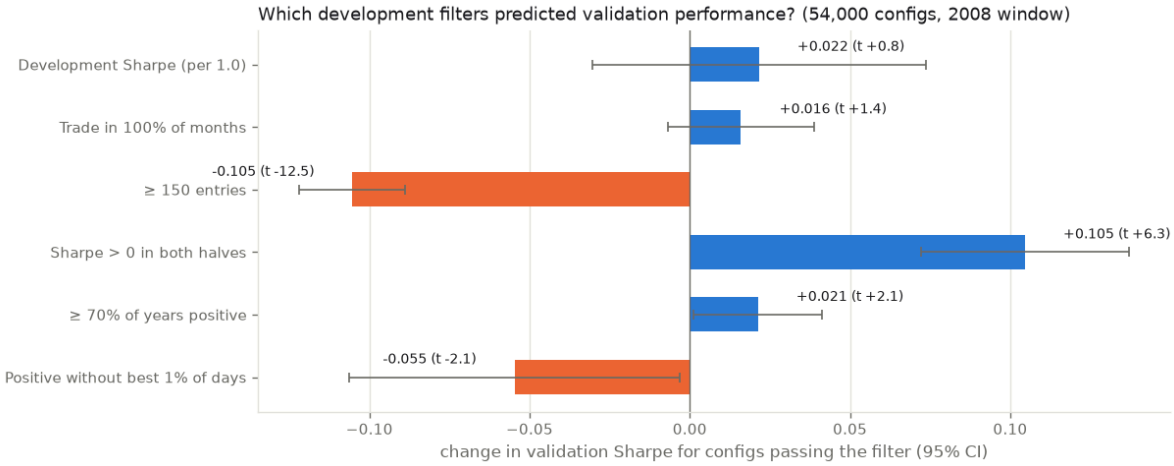


Median net Sharpe by factor set, common window (blue: development, orange: validation); ★ marks the original grid's three sets.

	dev median Sharpe	dev 90th pct Sharpe	val median Sharpe	val % > 0	val mean Sharpe of dev top 10%
factors					
TLT	0.16	0.67	-0.02	0.48	-0.07
SPY+TLT	0.14	0.67	-0.07	0.42	-0.16
IAU+GLD+GDXJ	0.12	0.49	-0.23	0.24	-0.29
GLD+GDXJ	0.11	0.50	-0.27	0.19	-0.31
IAU+GDXJ	0.10	0.49	-0.28	0.20	-0.31
GDXJ	0.09	0.48	-0.12	0.34	-0.25
SPY+IAU+GLD+TLT+GDXJ	0.08	0.49	-0.24	0.23	-0.39
SPY+IAU+GLD+GDXJ	0.06	0.43	-0.27	0.21	-0.32
SPY+GDXJ	0.06	0.46	-0.21	0.24	-0.20
SPY+TLT+GDXJ	0.06	0.44	-0.18	0.27	-0.18
IAU+GLD+TLT+GDXJ	0.05	0.46	-0.20	0.31	-0.20
SPY+GLD+TLT+GDXJ	0.05	0.50	-0.28	0.21	-0.41

8.3 Hard filters: removing them, then choosing which to keep

With all filters removed, ranking by neighbourhood score alone picked configurations with good neighbours but weak results of their own (L = 250, k = 1.25, H = 1); all ten shortlisted configurations then lost money in validation in both all-factors runs. To decide which filters are worth keeping, I measured whether passing each filter in development predicted better validation results across all 54,000 configurations of the 2008-window grid (one regression with all filters together, standard errors clustered by factor set × lookback):



Only "positive Sharpe in both halves" is a strong predictor; "≥ 150 entries" is strongly harmful (it favours low thresholds that trade on weak signals); development Sharpe itself predicts nothing.

	effect on validation Sharpe	std. error	t-stat
Development Sharpe (per 1.0)	0.022	0.027	0.811
Trade in 100% of months	0.016	0.012	1.373
≥ 150 entries	-0.105	0.008	-12.474
Sharpe > 0 in both halves	0.105	0.017	6.309
≥ 70% of years positive	0.021	0.010	2.073
Positive without best 1% of days	-0.055	0.026	-2.080

Filters kept: net Sharpe > 0 in both halves of development, ≥ 70% of calendar years positive, and at least one trade in every month (your requirement, and statistically neutral). **Dropped:** the 150-entry minimum and the "positive without the best 1% of days" gate (reported only).

8.4 Top configurations with the three kept filters

Ranked by development Sharpe; identical strategies (same trades under different policy/direction labels) are merged. Validation is shown for information only.

2008 window, 15 sets without GDXJ (996 of 54,000 pass):

factors	L	m	k	H	policy / direction	dev Sharpe	nbhd	entries	val Sharpe	val 1st half	val 2nd half
SPY+TLT	250	2	0.50	1	A, B, C · est	1.17	-0.13	1222	-0.05	-0.78	0.33
SPY+TLT	250	2	0.50	1	A, B, C · rev	1.12	0.29	1222	-0.05	-0.78	0.33
SPY+TLT	250	2	0.50	3	B · est	1.05	-0.13	574	-0.15	-1.11	0.38
TLT	250	2	0.50	1	A, B, C · est	1.04	-0.05	1233	0.02	-0.78	0.44
TLT	120	2	0.50	1	A, B, C · rev	1.04	0.29	1285	0.01	-0.38	0.23
SPY+TLT	250	2	0.50	10	B · est	1.02	-0.14	433	-0.11	-0.89	0.34
SPY+TLT	250	2	0.50	5	B · est	1.01	-0.14	463	-0.10	-0.85	0.33
SPY	250	2	0.50	1	A, B, C · est	1.01	-0.21	1230	-0.16	-0.43	-0.03

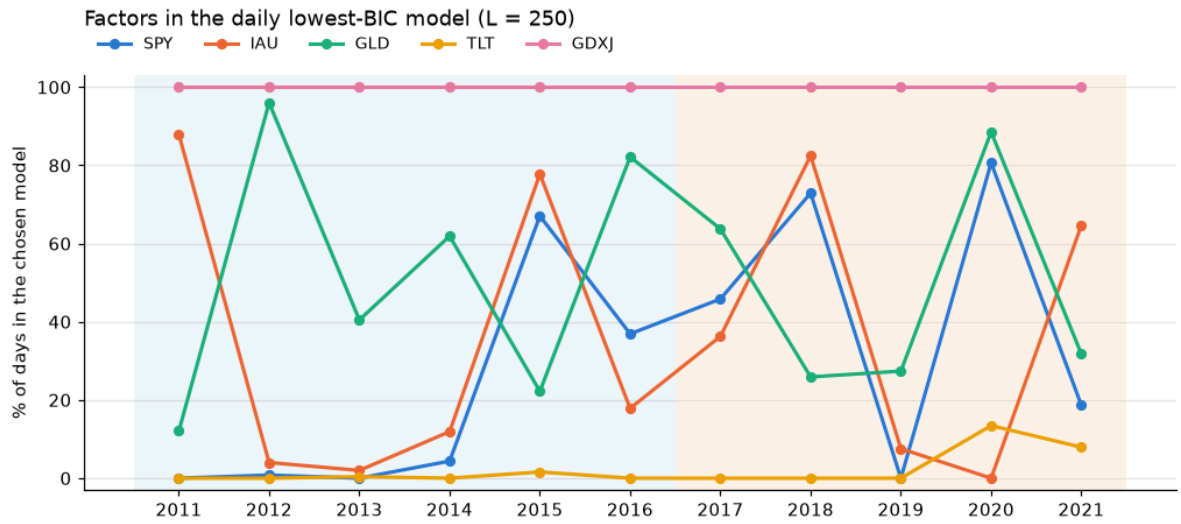
Common window, all 31 sets (2,264 of 111,600 pass):

factors	L	m	k	H	policy / direction	dev Sharpe	nbhd	entries	val Sharpe	val 1st half	val 2nd half
GLD+TLT	40	1	0.50	5	A · est	1.68	-0.07	231	0.42	1.44	-0.15
SPY+GDXJ	20	1	0.50	10	A · rev	1.31	-0.02	126	0.17	0.45	0.00
SPY+TLT+GDXJ	20	5	0.50	10	A · rev	1.30	-0.17	125	0.19	0.11	0.24
SPY+TLT	250	1	1.00	2	B · rev	1.29	0.77	331	-0.33	-0.29	-0.38
SPY+GDXJ	20	1	1.25	5	C · est	1.28	0.20	166	0.01	0.23	-0.13
TLT	60	2	0.50	1	A, B, C · rev	1.23	0.48	761	-0.24	-0.34	-0.20
TLT	40	2	0.50	1	A, B, C · rev	1.21	0.40	767	-0.33	-0.55	-0.20
TLT	120	2	0.50	1	A, B, C · est/rev	1.20	0.35	772	0.01	-0.38	0.23

The lists are dominated by TLT / SPY / SPY+TLT with m = 2, k = 0.5, H = 1 — effectively GDX's own two-day reversal — with development Sharpe ≈ 1.0–1.2 and validation ≈ 0. The one high validation number (GLD+TLT, L 40, k 0.5, H 5: 1.68 → 0.42) is an isolated peak (neighbourhood median −0.07) whose second half of validation was negative.

8.5 Best-fitting factor set chosen each day (BEST_BIC)

Each day, all 31 regressions are fitted on the trailing window and the one with the lowest BIC supplies that day's residual for the z-score and the forecast (no future data is used). The choice nearly always includes GDXJ:



Share of days each factor is in the chosen model, by year (L = 250).

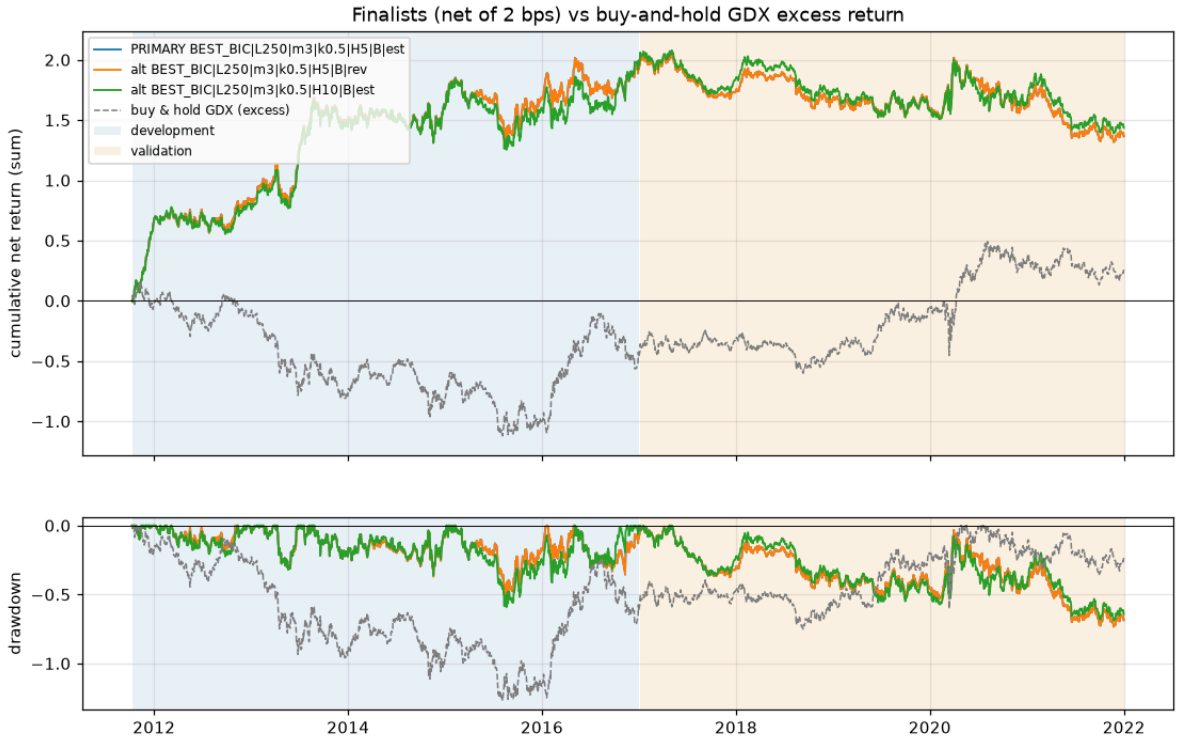
	share of days
chosen_set	
GLD+GDXJ	0.339
SPY+GLD+GDXJ	0.169
GDXJ	0.155
SPY+IAU+GDXJ	0.149
IAU+GDXJ	0.148
IAU+GLD+GDXJ	0.017

With the three kept filters, 53 of 3,600 configurations pass. The frozen pick, BEST_BIC with L 250, m 3, k 0.5, H 5, policy B, went from **0.88** Sharpe in development to **−0.32** in validation, losing in both halves. The yearly walk-forward looks better (stitched +0.45) but almost all of it comes from 2013–2014; its 2017–2021 years average −0.28.

Top BEST_BIC configurations (53 pass the filters):

factors	L	m	k	H	policy / direction	dev Sharpe	nbhd	entries	val Sharpe	val 1st half	val 2nd half
BEST_BIC	120	3	0.50	5	B · est/rev	0.98	0.38	228	-0.85	-0.60	-1.05
BEST_BIC	120	3	0.50	10	B · est/rev	0.94	0.38	213	-0.85	-0.74	-0.96
BEST_BIC	250	3	0.50	10	B · est/rev	0.89	0.54	216	-0.30	-0.61	-0.13
BEST_BIC	250	3	0.50	5	B · est/rev	0.88	0.46	235	-0.32	-0.55	-0.21
BEST_BIC	120	3	0.50	3	B · est/rev	0.83	0.31	266	-0.89	-0.60	-1.11
BEST_BIC	120	3	0.50	5	C · est/rev	0.82	0.39	186	-0.85	-0.83	-0.91
BEST_BIC	250	1	0.75	10	B · est	0.73	0.17	321	0.43	0.86	0.20
BEST_BIC	250	1	0.75	2	B · est	0.73	0.12	470	0.08	-0.07	0.17

Two configurations held up in validation (L 250, m 1, k 0.75, H 10 or 5, policy B: +0.43 and +0.30, positive in both halves), but they are 2 of 53 survivors after 169,200 configurations tried, with weak neighbourhoods (0.15–0.17). They are candidates for the out-of-sample test, not evidence of an edge.



BEST_BIC finalists vs buy-and-hold GDX (common window).

9. Interpretation

- **There is information in the residual** — placebo percentiles of 98–99.7%, consistently losing sign-flips, reversion beating the estimated direction in both periods, and a broad positive region in the heatmaps all say the residual's timing is not random.
- **But it is small relative to the search** — the DSR, the weak dev→validation rank correlation (0.20), the negative walk-forward and the tail dependence (kurtosis 17; losses once the top 1% of days are removed) all say the *selected* configuration's Sharpe is mostly selection luck on top of a modest true effect.
- **The model structure is mis-aimed** — the effect is a day-2 reversal while Stage 2 forecasts day 1; the estimated-direction configurations therefore underperform and the chosen ones rely on an imposed direction that Stage 2 does not endorse.
- **Changing the factors or the selection rules does not fix it** — single regression, all 31 factor combinations, no filters, data-chosen filters and a daily best-fit model all show the same collapse from development to validation. A better-fitting factor model explains more of today's move but does not make the leftover residual more predictable.
- **The validation period was hard for this whole family** (grid median −0.27 vs buy-and-hold GDX +0.41), so part of the failure is regime, not only overfitting.

10. Next steps and possible changes

> Any change below is made **after** seeing validation, so it must be logged as a protocol deviation and judged only on data that has not been used — the coming OOS file (or a later period). The frozen configuration should stay untouched and be evaluated on OOS first, as the honest test of what was built.

9.1 Immediately

1. **Run the frozen primary on OOS unchanged** with `run_oos.py`; report it against its flag (it failed the validation filters) and against buy-and-hold GDX. Treat it as a test, not a deployment.
2. **Record the frozen hash externally** (`6981dc95…`) so the OOS run can be checked with `--expected-sha256` .

9.2 Model changes suggested by the evidence (each needs fresh data to judge)

1. **Fix the horizon mismatch.** Make Stage 2 forecast the 2-day return ($ExRet_{t+1}$ + $ExRet_{t+2}$) or regress on lagged residuals (e_t, e_{t-1}) jointly, so the estimated direction can capture the day-2 reversal instead of relying on an imposed sign.
2. **Use position sizing instead of a binary gate.** Scale exposure by $|\hat{r}|$ / forecast volatility (or target a fixed volatility), capped at 1x. This addresses the kurtosis / tail-day dependence that failed the "ex top-1% days" filter more directly than tuning k.
3. **Guarantee monthly activity differently.** The strongest near-misses (k = 1.25, L = 250) fail only the monthly rule. A quantile-based threshold (trade the top q% of |z| within a rolling month) or a "at least one trade per month, taking the largest |z|" fallback would keep the requirement while allowing a higher-conviction threshold.
4. **Ensemble instead of a single winner.** Average positions across a pre-declared neighbourhood (e.g. all L × k cells at H = 2, reversion, m = 1) rather than picking one cell; this directly targets the walk-forward instability.
5. **Execution realism.** Signal and execution at the same close is optimistic (market-on-close orders are submitted before the close). Test signal at ≈ 15:50 prices or execution at the next close; the day-2 location of the edge suggests a one-day delay may cost less than expected. Add short-borrow cost and a slippage model beyond 2 bps.
6. **Richer residual model.** Consider miners-specific drivers (oil/energy as a cost proxy, USD, equity-sector factors) or GDXJ–GDX spread information; test whether the residual reversal is stronger after large GLD moves or on high-volume days.

9.3 Protocol improvements

1. **Smaller, pre-registered grid.** 10,800 trials made the DSR hurdle ≈ 1.4 annualised Sharpe. Fixing dimensions that the evidence already settles (direction = reversion, H ≈ 2) and searching only a few others would lower the hurdle legitimately.
2. **Effective number of trials.** Estimate it from the correlation structure of the 10,800 return series (e.g. clustering or eigenvalue count) and report DSR at that N alongside N = 10,800.
3. **Longer holdouts.** With a thin effect, 5 years of validation cannot separate Sharpe 0.3 from 0; plan the next evaluation to use the full OOS period plus any later data, with nothing re-tuned on it.
4. **Cluster-robust event-study bands.** The current ±1 s.e. bands treat overlapping events as independent and are too narrow.

11. Reproducing everything

```
cd gdx_strategy
pip install -r requirements.txt
python -m pytest -q                     # 83 tests
python scripts/run_is.py                # development stage only (no validation access)
python scripts/run_is.py --report       # rebuild report.html / grid_results.parquet from the frozen config
python scripts/run_oos.py --is data/raw/gold_etf_factor_regression.xlsx --oos data/raw/<OOS>.xlsx --out results/oos/
python scripts/run_is.py --grid config/grid_best_bic.yaml --results results/variants/best_bic --validate --reason "..."
python scripts/factor_combinations.py   # 31-combination comparison
python scripts/top_configs.py           # top configs after the three kept filters
python scripts/param_cube.py            # interactive 3-D explorer
```

Key outputs in `results/` : frozen_config.json (+ .sha256), report.html, dev_shortlist.csv, validation_shortlist.csv, walk_forward.csv, all_configs_summary.csv.gz (every configuration × period), validation_log.jsonl (every validation access, including the post-freeze reporting entry), figures/.